

HEALTH & MEDICAL DATA SCIENCE PROJECT

Maternal Health Risk Stratification

Statistical Analysis, Exploratory Data Analysis, and Interpretable Machine Learning on the UCI Maternal Health Risk Dataset

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n = 1,012 patients · Python · SQL · scikit-learn

AGENDA

What we'll cover

A reproducible workflow from raw sensor data to an interpretable screening model.

01

Introduction & Data

Research question, source, and cleaning decisions

02

Exploratory Analysis

Risk groups, Blood Sugar, BP, Age & Heart Rate

03

SQL & Python Pipeline

Reproducible queries and preprocessing

04

Machine Learning

Logistic Regression vs. Decision Tree

05

Results & Dashboards

Metrics, feature importance, BI visuals

06

Limitations & Next Steps

Scope, caveats, and recommendations

INTRODUCTION

Why Maternal Risk Stratification?

Digital health systems and low-cost IoT sensors let us monitor blood pressure, glucose, temperature, and heart rate at scale. The UCI Maternal Health Risk dataset was collected from hospitals and community clinics in rural Bangladesh through such a system.

RESEARCH QUESTION

How strongly do a small set of routine physiological variables distinguish Low, Mid, and High maternal risk — and can an interpretable baseline classifier identify High-Risk patients with useful sensitivity?

1,012

patients analyzed

6

physiological variables

3

risk classes: Low / Mid / High

Data Source & Variables

UCI Maternal Health Risk dataset (Dataset 863) — Ahmed (2020), DOI 10.24432/C5DP5D. Collected via an IoT-based risk-monitoring system in rural Bangladesh; 1,014 raw observations, no missing values.

Variable	Type / Unit	Description
Age	Numeric; years	Maternal age recorded in years
SystolicBP	Numeric; mmHg	Systolic blood pressure
DiastolicBP	Numeric; mmHg	Diastolic blood pressure
BS	Numeric; mmol/L	Blood sugar (glucose) concentration
BodyTemp	Numeric; °F	Body temperature
HeartRate	Numeric; bpm	Heart rate
RiskLevel	Categorical (target)	Low Risk, Mid Risk, or High Risk

Data Cleaning & Quality Decisions

0

Missing values

None found — no imputation needed.

562

Exact duplicates

Retained. Coarse sensor rounding can legitimately create identical rows across different patients.

2

Implausible Heart Rate

Two records with HR = 7 bpm removed as sensor/entry errors — no replacement imputed.

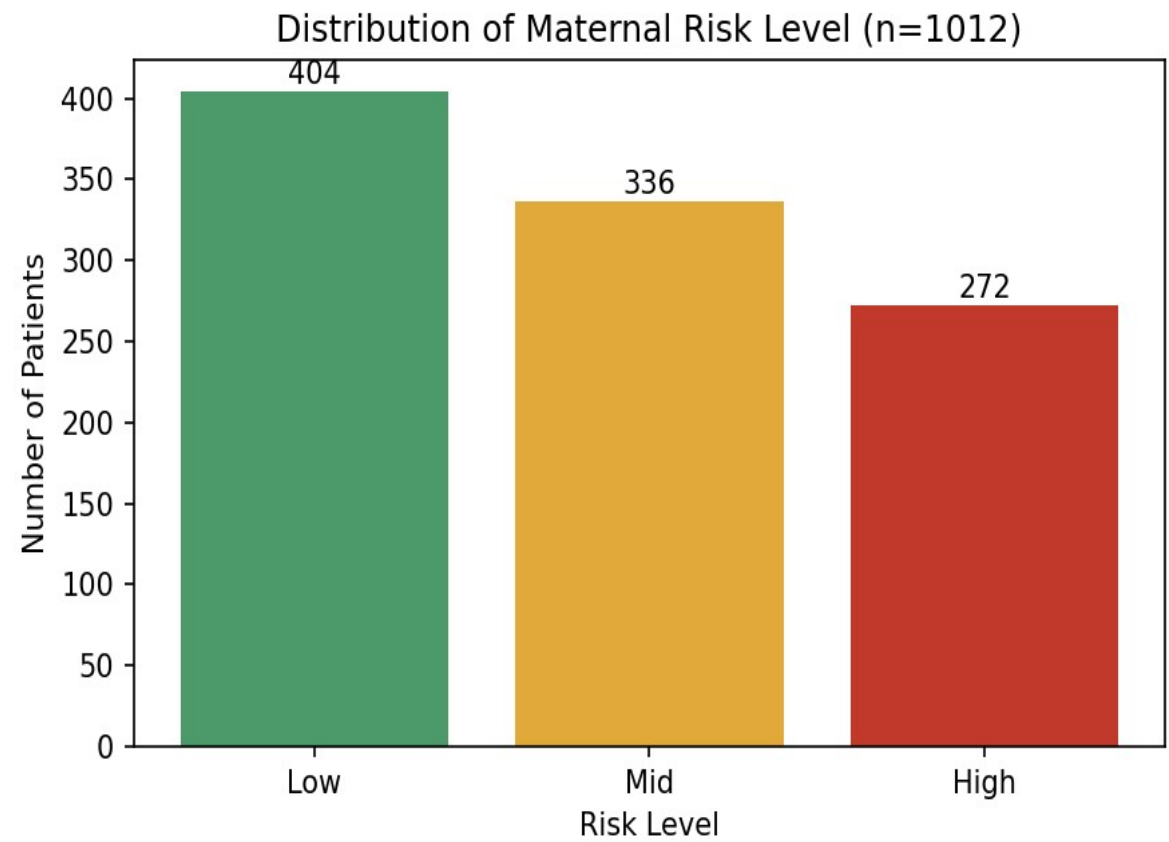
1,012

Final analytical sample

Used consistently across every table, figure, model, and dashboard in this deck.

Final class balance: Low Risk 404 (39.9%) • Mid Risk 336 (33.2%) • High Risk 272 (26.9%)

Risk-Level Distribution

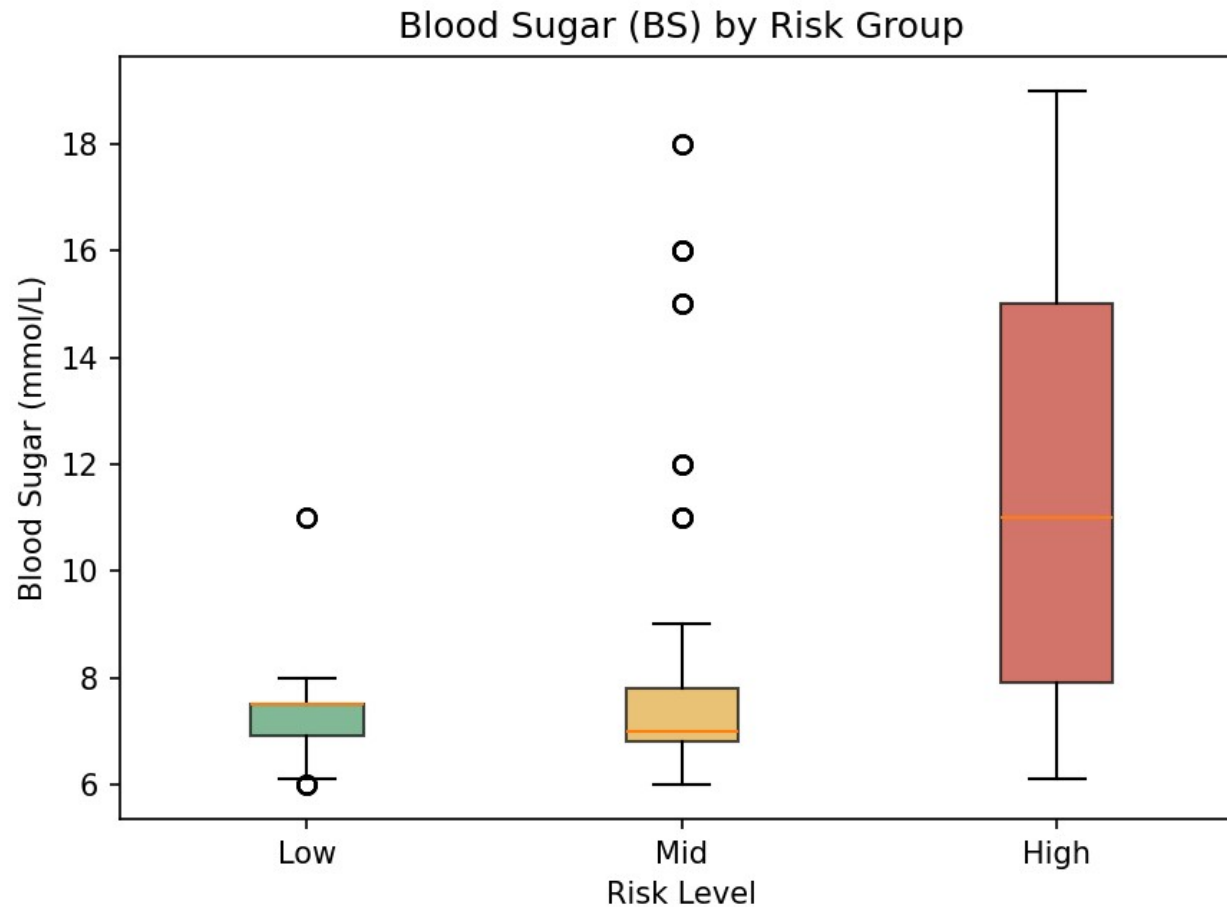


Group-level means

Risk	n	Age	SBP	DBP	BS	HR
Low	404	26.9	105.8	72.5	7.22	73.1
Mid	336	28.4	113.2	74.2	7.80	74.2
High	272	36.2	124.2	85.1	12.12	76.7

High-Risk patients are older, and show meaningfully higher blood pressure and blood sugar than Low-Risk patients. Not an extreme class imbalance, but large enough that accuracy alone won't tell the full story.

Blood Sugar — The Strongest Signal

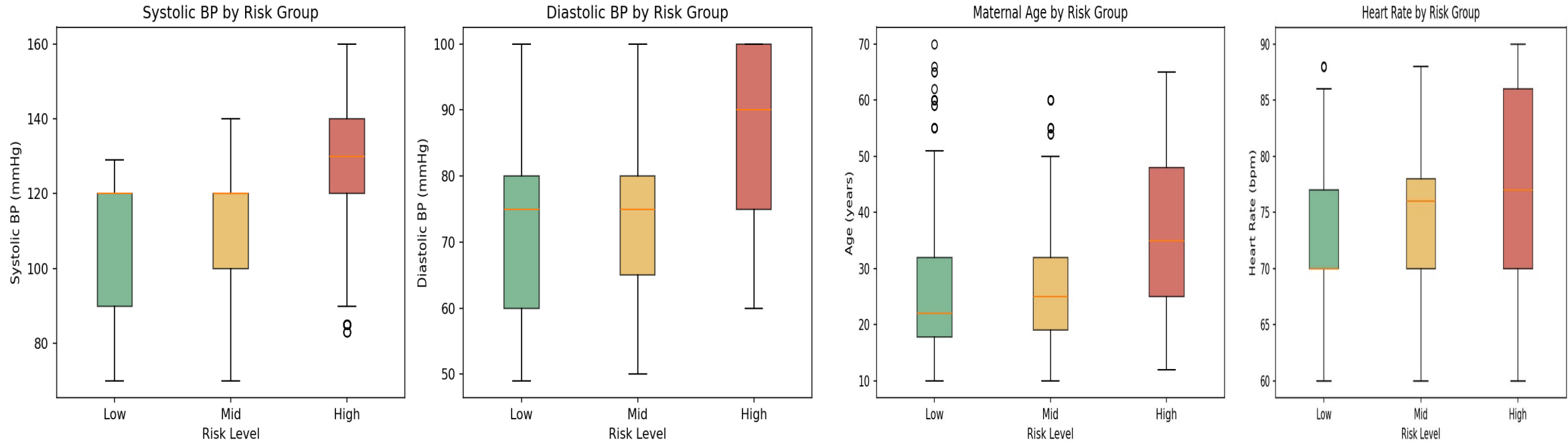


12.12 mmol/L

mean Blood Sugar in High Risk vs. 7.22 mmol/L in Low Risk

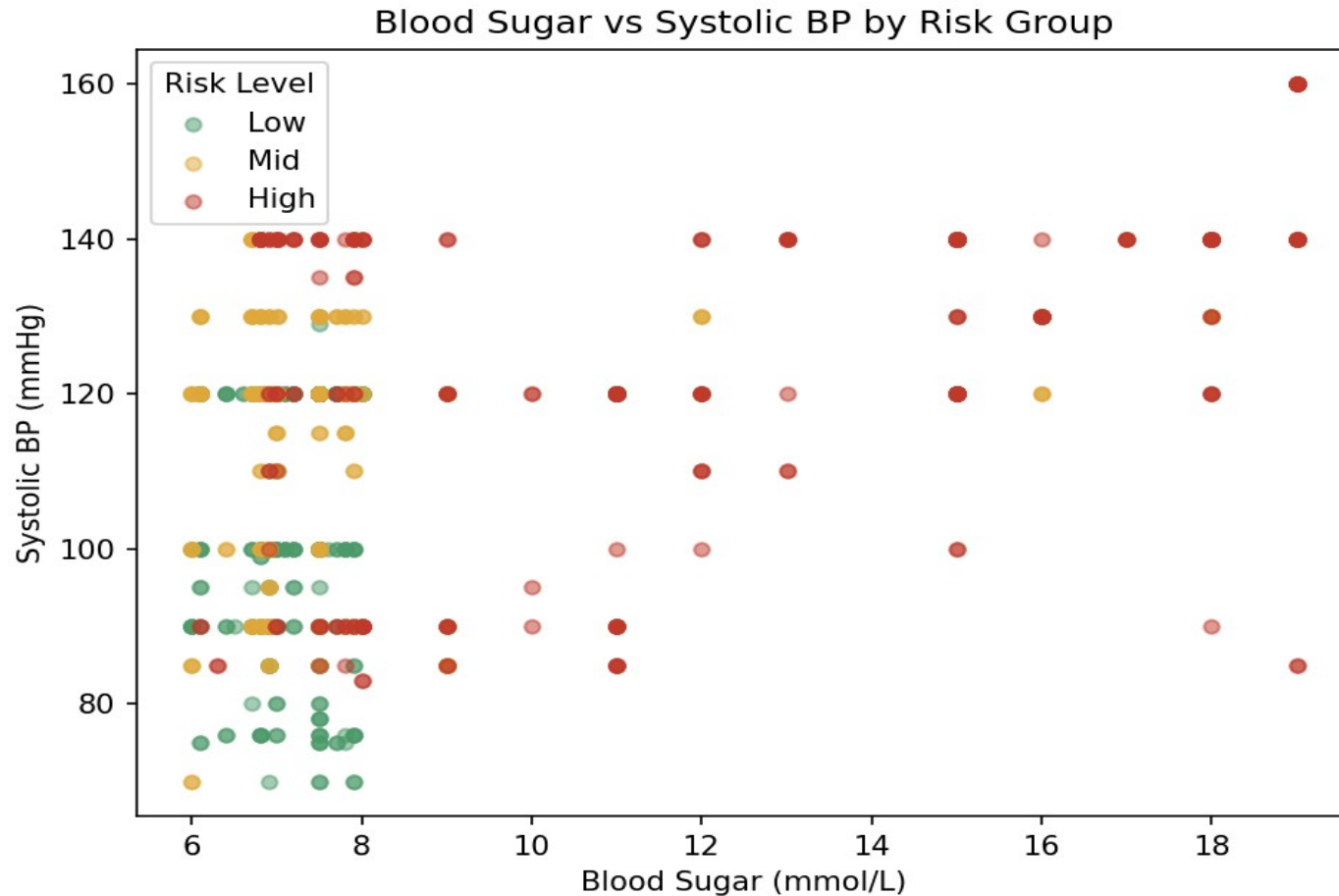
- Threshold rule (BS ≥ 11 mmol/L): met by 62.5% of High-Risk patients, vs. only 1.0% of Low-Risk and 9.5% of Mid-Risk.
- Standard deviation widens sharply with risk: 0.64 (Low) $\rightarrow$ 2.28 (Mid) $\rightarrow$ 4.17 (High).
- This shows strong statistical association — not proof that blood sugar causes the assigned risk label.

Blood Pressure, Age & Heart Rate



Systolic BP rises from 105.8 mmHg (Low) to 124.2 mmHg (High); diastolic from 72.5 to 85.1 mmHg — but Low and Mid overlap substantially. Age shifts upward (26.9 → 36.2 years) yet stays heterogeneous. Heart Rate shows the weakest separation of any variable (73.1 / 74.2 / 76.7 bpm) — no single variable is a perfect classifier on its own.

Joint Relationship: Blood Sugar × Systolic BP



- High-Risk points concentrate toward higher values of both Blood Sugar and Systolic BP.
- No clean boundary separates the classes — a bivariate view still leaves overlap.
- The RiskLevel label may itself be partly derived from BP/BS thresholds, so a model built on these features may partly reproduce the labeling rule rather than discover new clinical mechanisms.

METHODS

Reproducible Analytical Pipeline

Implemented in Python, supplemented by a SQLite SQL layer — same 1,012-row dataset, two ways.

- 1 Load UTF-8 CSV & normalize categorical labels
- 2 Audit missingness, duplicates, implausible values
- 3 Remove only the 2 implausible HeartRate rows
- 4 Compute descriptive statistics by RiskLevel
- 5 Produce distribution, boxplot & scatter figures
- 6 80/20 stratified train/test split (random_state=42)
- 7 Fit StandardScaler on training data only
- 8 Train multinomial Logistic Regression + Decision Tree
- 9 Evaluate via confusion matrices, precision/recall/F1
- 10 Select final model: High-Risk Recall, then macro F1

Reproducibility with SQLite

Nine documented queries reproduce the class distribution, group summaries, and threshold checks directly against the cleaned 1,012-row table — no pandas required.

```
-- Group summaries: mean of every clinical variable by RiskLevel
SELECT
  RiskLevel,
  COUNT(*)                AS n,
  ROUND(AVG(Age), 1)       AS avg_age,
  ROUND(AVG(SystolicBP), 1) AS avg_systolic_bp,
  ROUND(AVG(BS), 2)        AS avg_blood_sugar
FROM maternal_health
GROUP BY RiskLevel
ORDER BY
  CASE RiskLevel
    WHEN 'low risk' THEN 1
    WHEN 'mid risk' THEN 2
    WHEN 'high risk' THEN 3 END;
```

Threshold finding (Q3a)

Risk	n	BS ≥ 11 mmol/L
Low	404	1.0%
Mid	336	9.5%
High	272	62.5%

SQLite has no native STDDEV or MEDIAN — both are reconstructed with window functions and $\sqrt{E[X^2] - E[X]^2}$, matching the Python descriptive statistics exactly.

Leakage-Safe Split & Logistic Regression

```
FEATURES = ["Age", "SystolicBP", "DiastolicBP",  
            "BS", "BodyTemp", "HeartRate"]  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2,  
    random_state=42, stratify=y  
)  
  
# Fit scaler on TRAIN ONLY to avoid leakage  
scaler = StandardScaler().fit(X_train)  
X_train_s = scaler.transform(X_train)  
X_test_s = scaler.transform(X_test)  
  
logreg = LogisticRegression(max_iter=2000,  
                             random_state=42)  
logreg.fit(X_train_s, y_train)
```

Multinomial Logistic Regression

- 80/20 stratified split preserves the Low/Mid/High proportions in both sets (test n = 203, including 55 High-Risk cases).
- Softmax over per-class linear scores $z_k = \beta_{0k} + \beta_{1k}X_1 + \dots + \beta_{pk}X_p$ converts scores to class probabilities.
- Standardization matters here because predictors are on very different scales (mmHg vs. mmol/L vs. °F).
- Result: 0.586 accuracy, 0.584 macro F1, 0.782 High-Risk Recall.

Shallow, Interpretable Decision Tree

```
tree = DecisionTreeClassifier(  
    max_depth=4,  
    min_samples_leaf=15,  
    random_state=42  
)  
tree.fit(X_train, y_train)          # no scaling needed  
y_pred = tree.predict(X_test)  
  
# Select final model: High-Risk Recall first,  
# macro F1 as tie-breaker  
if tree_metrics["high_risk_recall"] >= \  
    lr_metrics["high_risk_recall"]:  
    final_model = tree
```

$$Gini(t) = 1 - \sum p_k^2$$

Decision Tree (depth = 4)

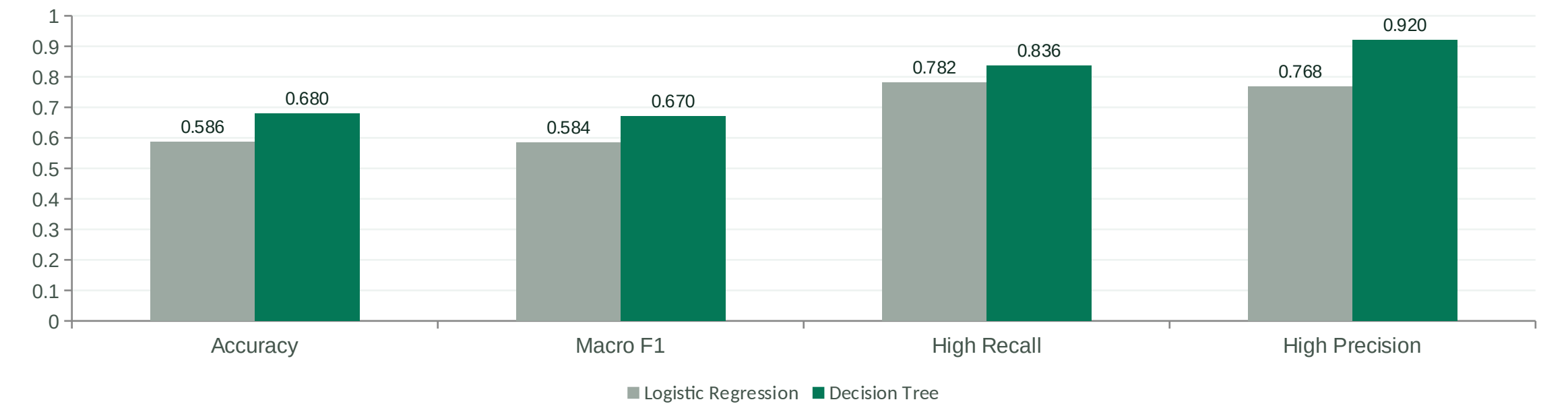
- Kept intentionally shallow (depth 4, min leaf 15) to stay interpretable and limit overfitting on a modest sample.
- Each split reduces Gini impurity — the tree can be read top-to-bottom as an explicit set of threshold rules.
- Chosen as the final model: it outperformed Logistic Regression on all four reported metrics.
- Result: 0.680 accuracy, 0.670 macro F1, 0.836 High-Risk Recall, 0.920 High-Risk Precision.

RESULTS

Model Comparison on the Held-Out Test Set

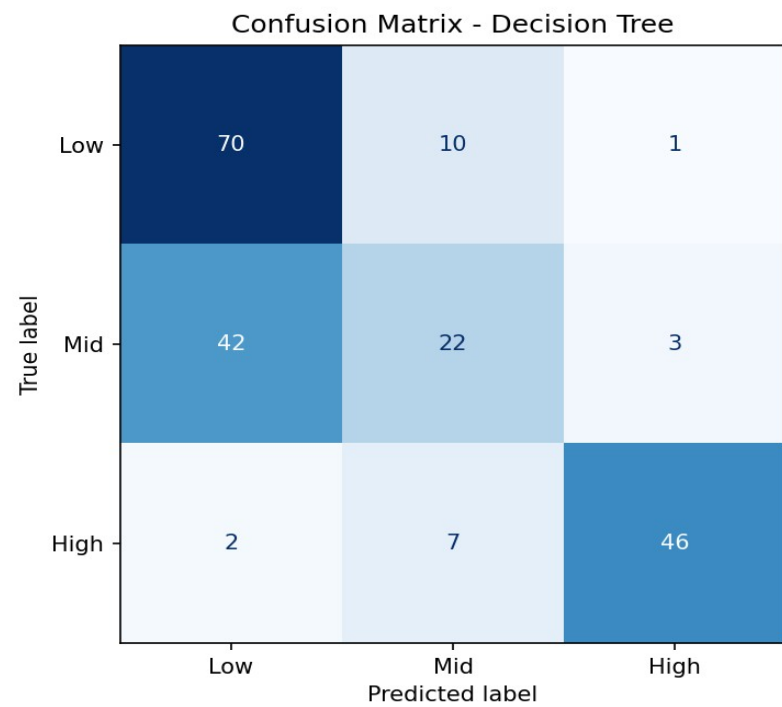
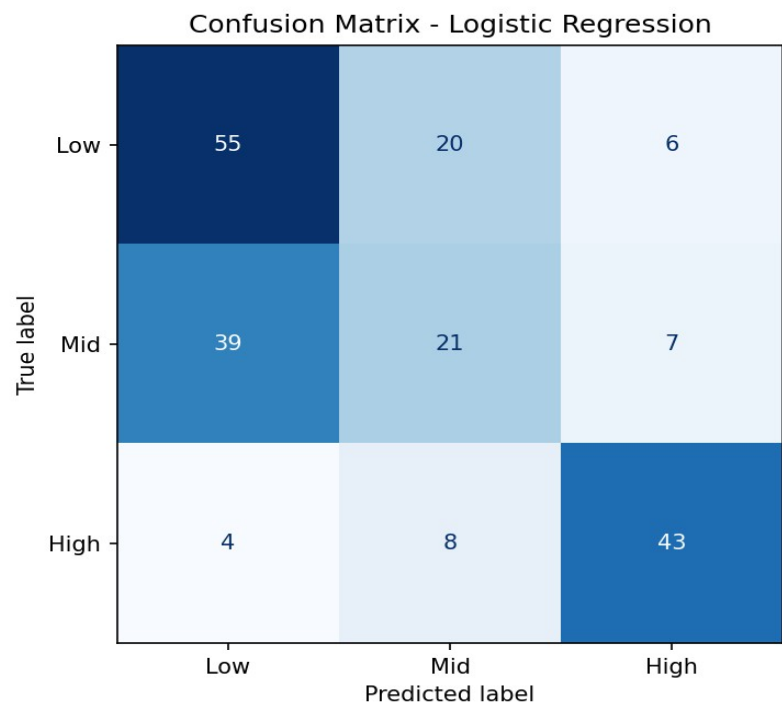
Model	Accuracy	Macro F1	High Recall	High Precision
Logistic Regression	0.586	0.584	0.782	0.768
Decision Tree — selected	0.680	0.670	0.836	0.920

Metric-by-Metric Comparison



RESULTS

Class-Wise Performance & Confusion Matrices

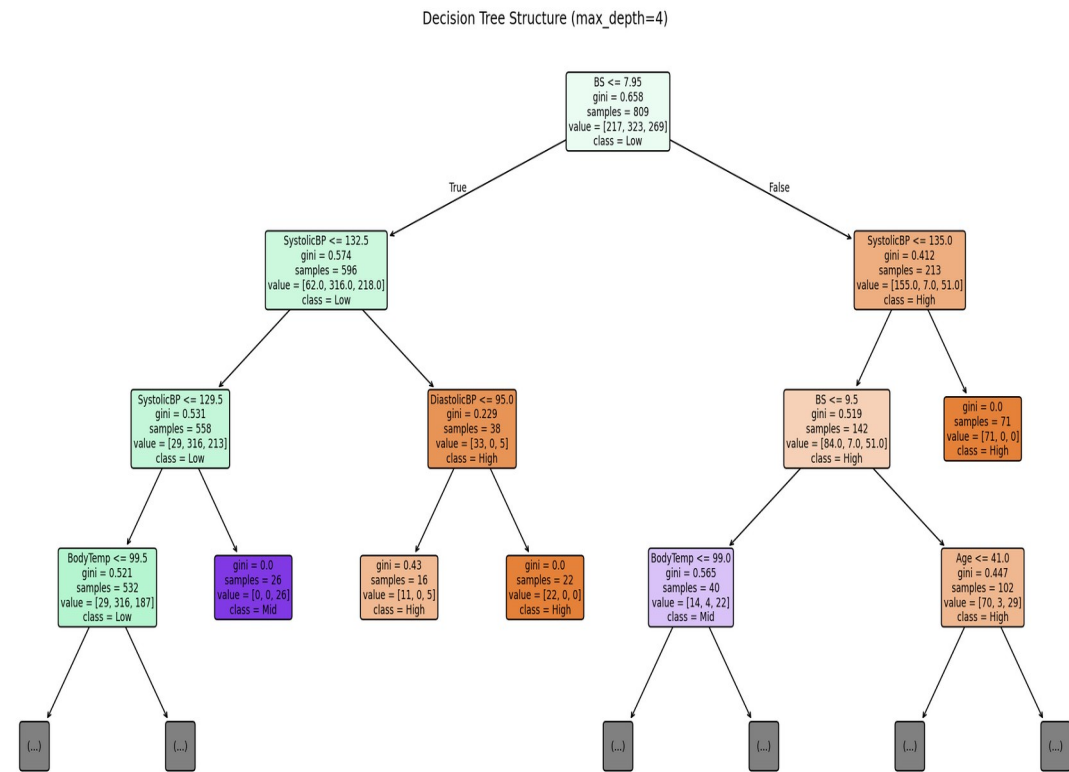
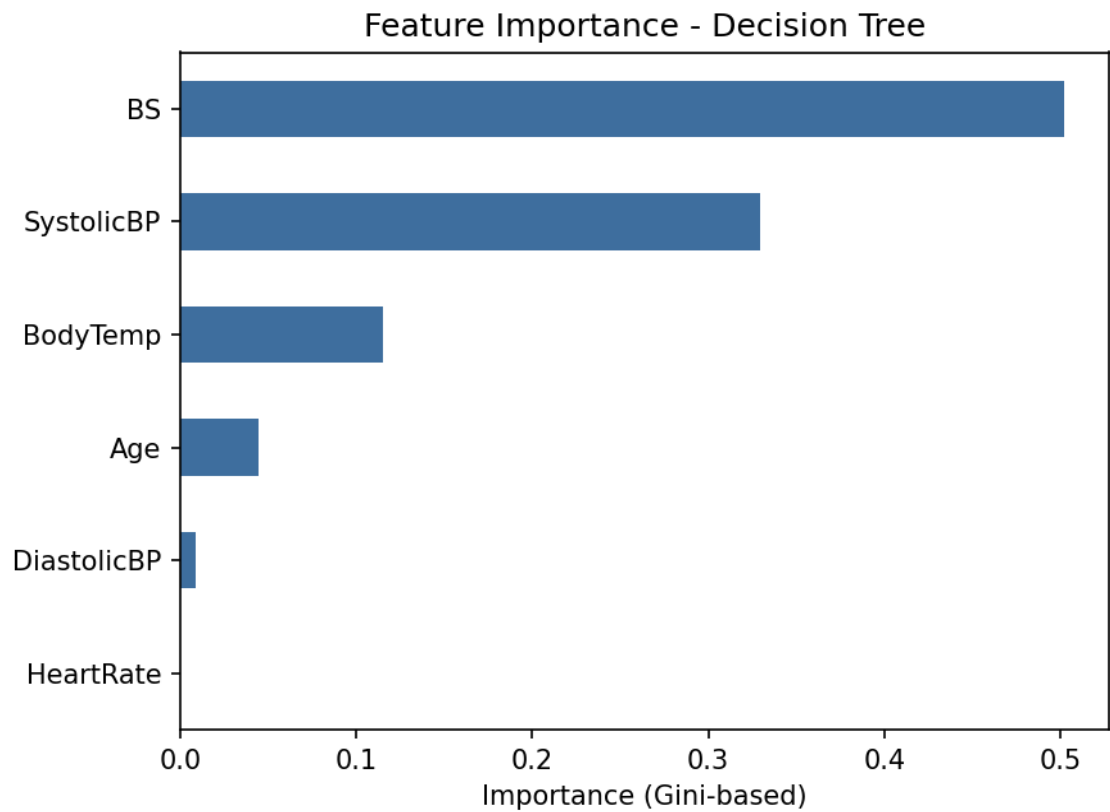


Decision Tree, per class

Class	Precision	Recall	F1
Low Risk	0.614	0.864	0.718
Mid Risk	0.564	0.328	0.415
High Risk	0.920	0.836	0.876

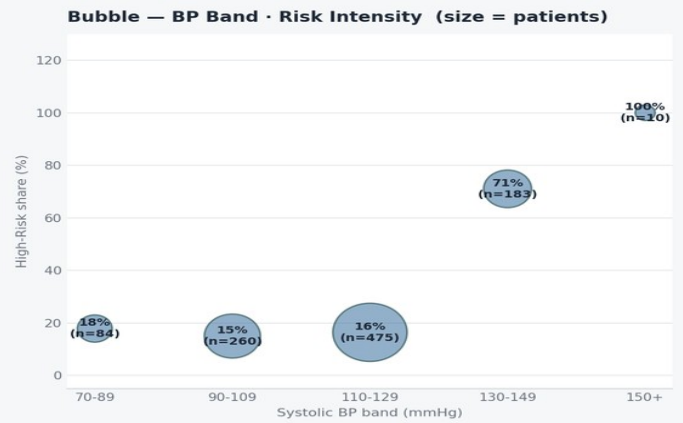
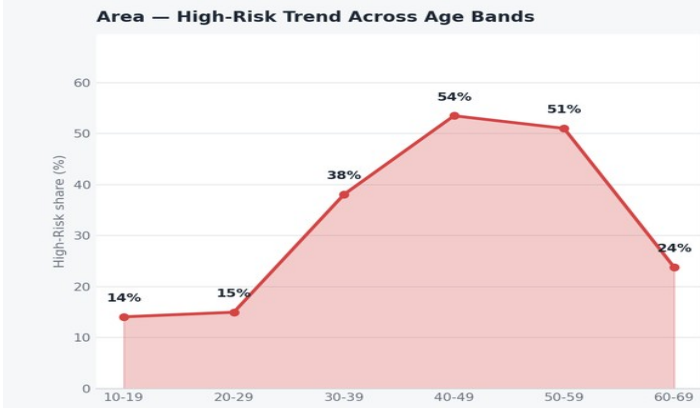
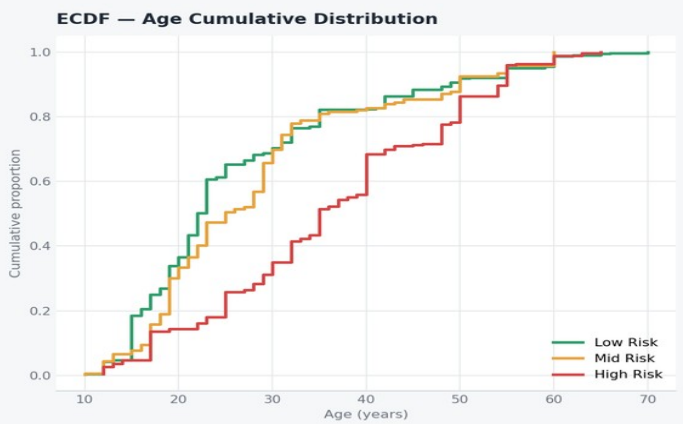
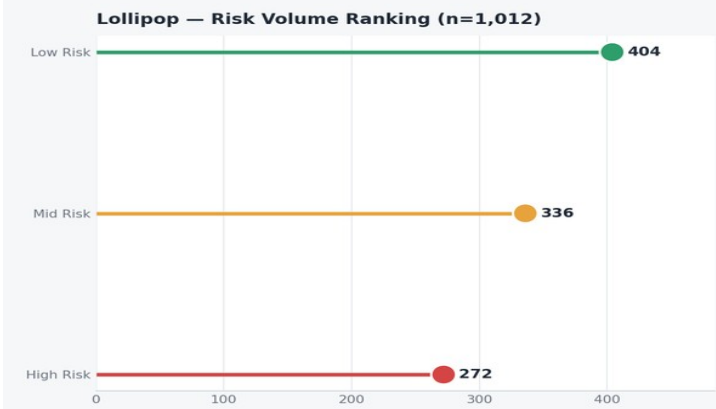
Main error: 42 of 65 Mid-Risk cases predicted as Low Risk — the Low/Mid boundary, not High-Risk detection, is the model's weak spot. Still, 9 of 55 true High-Risk cases were missed.

What Drives the Model?

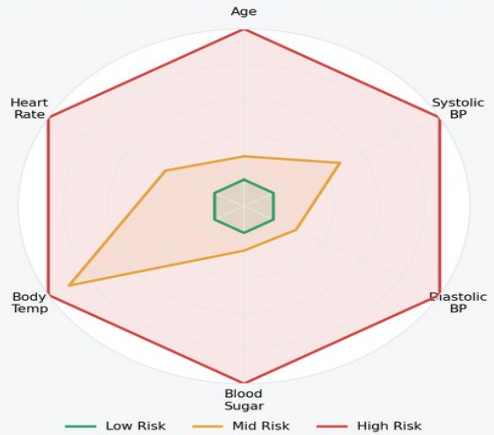


Blood Sugar (~50%) and Systolic BP (~33%) account for the large majority of the tree's splitting power — fully consistent with the descriptive EDA. Diastolic BP, Age, and Heart Rate contribute far less. Importance reflects impurity reduction, not a causal effect.

Overview Dashboard

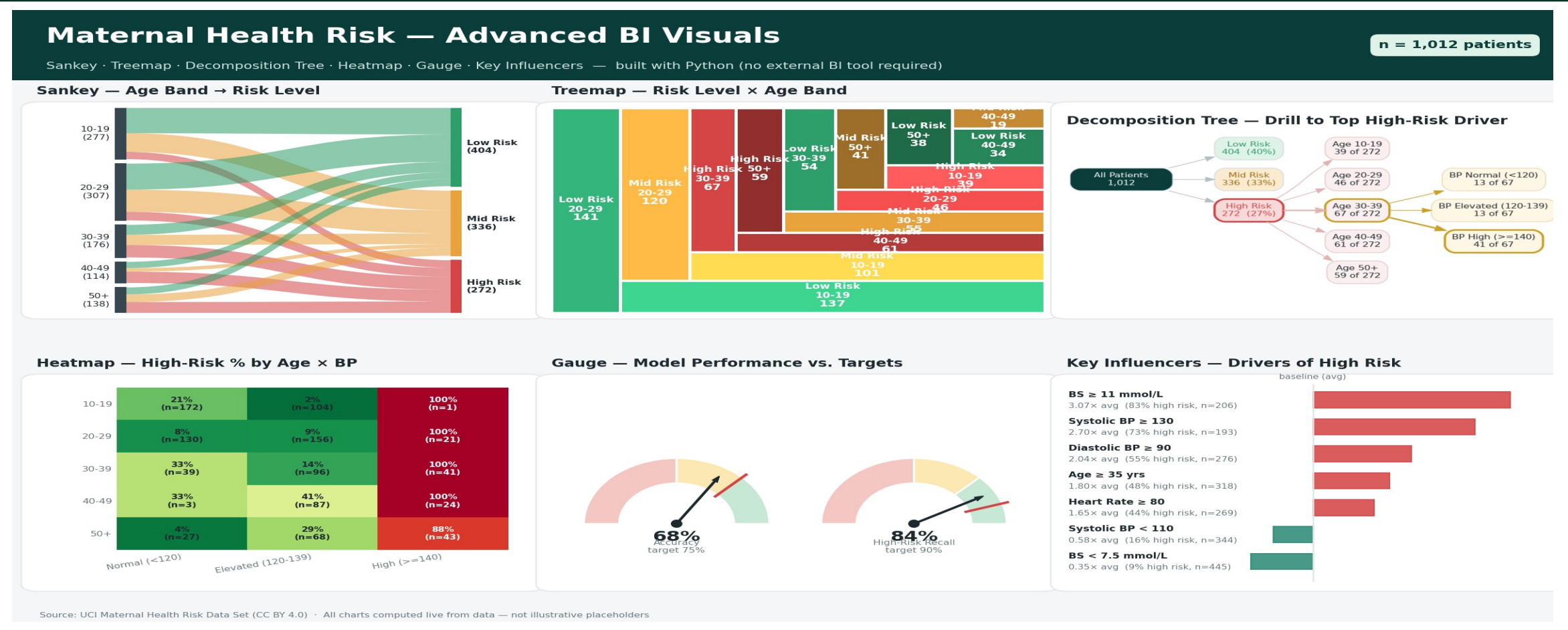


Radar — Clinical Profile Index (normalized mean)



Advanced BI Visuals

Sankey · Treemap · Decomposition Tree · Heatmap · Gauge · Key Influencers — built with Python, no external BI tool required.



DISCUSSION

Limitations

- **Geographic scope**

Rural Bangladesh only — may not generalize to other health systems.

- **Duplicate rows**

562 exact duplicates from coarse rounding reduce effective sample diversity.

- **No external validation**

Evaluated on one held-out split, not an independent cohort.

- **Limited predictors**

Only 6 fields; no gestational age, obstetric history, labs, or comorbidities.

- **Possible label dependence**

RiskLevel may already be partly derived from BP/BS thresholds.

- **Small High-Risk test set**

Only 55 High-Risk cases in the test split — recall estimates carry sampling variability.

CONCLUSION

A Transparent Baseline, Not a Clinical Decision Tool

Blood Sugar and blood pressure emerged as the dominant, consistent drivers across statistics, SQL, and machine learning alike. The shallow Decision Tree reached 0.836 High-Risk Recall and 0.920 High-Risk Precision — a strong screening pattern for an interpretable baseline. Future work should add clinically validated outcomes, larger and more diverse cohorts, cost-sensitive evaluation, and external validation before any real-world use.

THANK YOU

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Educational research project — results are not intended for clinical diagnosis or decision-making.